MATERIAL SAFETY DATA SHEET

SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

1.1. Product identifier

Product name: Athen Marine REF N Cool oil 68

Container size: -

1.2. Relevant identified uses of the substance or mixture and uses advised against

Application: Marine Refrigeration Oil

1.3. Details of the supplier of the safety data sheet

Supplier: Athen Marine Lubricants LLC
PO Box: 111541
Dubai, UAE.
Tel: +971 4 258 7297
www.athenmarine.com

Emergency contact: Tel: +971 50 483 4180

SECTION 2: HAZARDS IDENTIFICATION

2.1. Classification of the substance or mixture

Product definition Mixture Classification according to Regulation (EC) No. 1272/2008 [CLP/GHS] Asp. Tox. 1, H304 The product is classified as hazardous according to Regulation (EC) 1272/2008 as amended. See Section 16 for the full text of the H statements declared above.

2.2. Label elements

H 304: May be fatal if swallowed and enters the airways.

P301 + P310 + P331 - IF SWALLOWED: Immediately call a POISON CENTER or physician. Do NOT induce vomiting

P405 - Store locked up

P501 - Dispose of contents/container by all local, regional, national, and international regulations.

Annex XVII - Restrictions on the manufacture, placing on the market and use of certain dangerous substances, mixtures and articles – Not applicable

2.3. Other hazards

Other: Prolonged or repeated contact with skin may cause redness, itching, irritation, eczema, skin cracking and oil acne. The harmful effects may increase in used oil.

Hazard statement(s)
Not Applicable.

Precautionary statement(s) Prevention
Not Applicable

Precautionary statement(s) Response
Not Applicable

Precautionary statement(s) Storage: Not Applicable
Not Applicable

Not classified as Dangerous Goods by the criteria of the "Australian Code for the Transport of Dangerous Goods by Road & Rail" and the "New Zealand NZS5433: Transport of Dangerous Goods on Land"

SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

3.1 Mixtures

Annex I Nota L applies to the base oil(s) in this product. Nota L - The classification as a carcinogen need not apply if it can be shown that the substance contains less than 3 % DMSO extract as measured by IP 346.

Ingredient Name	% (w/w)	CAS Number
Severely Hydro treated Heavy Naphthenic Petroleum Oil	99	CAS : 64742-52-5
(IP 346 DMSO extract). Proprietary performance additives	<3	-

There are no additional ingredients present which, within the current knowledge of the supplier and in the concentrations applicable, are classified as hazardous to health or the environment, are PBTs or vPvBs or have been assigned a workplace exposure limit and hence require reporting in this section

Type

[1] Substance classified with a health or environmental hazard [2] Substance with a workplace exposure limit
[3] Substance meets the criteria for PBT according to Regulation (EC) No. 1907/2006, Annex XIII
[4] Substance meets the criteria for vPvB according to Regulation (EC) No. 1907/2006, Annex XIII
[5] Substance of equivalent concern

SECTION 4: FIRST AID MEASURES

4.1. **Description of first aid measures**

Inhalation: Move into fresh air and keep at rest. In case of persistent throat irritation or coughing: Seek medical attention and bring these instructions.

Skin contact: Remove contaminated clothing. Rinse skin immediately. Thoroughly rub with liquid soap and wash with lukewarm water. Get medical attention if any discomfort continues.

Eye contact: Immediately flush with plenty of water for up to 15 minutes. Remove any contact lenses and open eyelids widely. If irritation persists: Seek medical attention and bring along these instructions.

Ingestion: DO NOT induce vomiting. Get medical attention immediately.

4.2. **Most important symptoms and effects, both acute and delayed**

Symptoms/effects: See section 11 for more detailed information on health effects and symptoms.

4.3. **Indication of any immediate medical attention and special treatment needed**

Medical attention/treatments: If there is any suspicion of aspiration into the lungs either directly or as a result of vomiting, obtain medical advice.

SECTION 5: FIREFIGHTING MEASURES

5.1. **Extinguishing media**

Extinguishing media:

Extinguish with foam, carbon dioxide, dry powder or water fog. Cool containers exposed to heat with water spray and remove container, if no risk is involved.

Do not use water jet as an extinguisher, as this will spread the fire.

5.2. **Special hazards arising from the substance or mixture**

Specific hazards: When heated and in case of fire, harmful vapours/gases may be formed. Fire in closed areas should only be extinguished by trained personnel.

5.3. **Advice for firefighters**

Protective equipment for fire-: Wear suitable respiratory protective equipment.

fighters:

SECTION 6: ACCIDENTAL RELEASE MEASURES

6.1. Personal precautions, protective equipment and emergency procedures

Personal precautions: Mark the spillage. Avoid inhalation of oil mist and contact with skin and eyes. Keep public away from danger area. In case of spills, beware of slippery floors and surfaces.

Protective equipment: Wear necessary protective equipment.

6.2. Environmental precautions

Environmental precautions: Avoid discharge into drains. Contact local authorities in case of spillage to drain/aquatic environment.

6.3. Methods and material for containment and cleaning up

Methods for cleaning up: Dam and absorb spillages with sand, earth or other non-combustible material.

6.4. Reference to other sections

Reference: For personal protection, see section 8. For waste disposal, see section 13.

SECTION 7: HANDLING AND STORAGE

7.1. Precautions for safe handling

Safe handling advice: Observe good chemical hygiene practices. Avoid inhalation of oil mist and contact with skin and eyes. Avoid prolonged and repeated contact with oil, particularly used oil. Wash hands before breaks and before smoking, eating or drinking.

Technical measures: Use work methods which minimize oil mist production. Do not smoke or use open fire or other sources of ignition.

Technical precautions: When working with heated oil, mechanical ventilation may be required.

7.2. Conditions for safe storage, including any incompatibilities

Technical measures for safe storage: Store in tightly closed original container in a well-ventilated place.

Storage conditions: Keep at a temperature not exceeding: 45°C.

7.3. Specific end use(s)

Specific use(s): Lubricant.

SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

8.1. Control parameters

Occupational exposure limits:

CAS-No.:	Chemical name:	As:	Exposure limits:	Type:	Notes:	References:
-	Oil mist, - mineral	-	5 mg/m3	PEL (Long Term)	-	
		-	10 mg/m3	PEL (Short Term)	-	

8.2. Exposure controls

Engineering measures: Provide adequate ventilation. Observe Occupational Exposure Limits and

	minimize the risk of inhalation of vapours and oil mist.
Respiratory equipment:	No specific recommendations.
Hand protection:	Gloves (synthetic rubber, neoprene)
Eye protection:	Wear approved, tight fitting safety glasses where splashing is probable.
Skin protection:	Wear appropriate clothing to prevent repeated or prolonged skin contact.
Hygiene measures:	Wash hands after handling. Observe good chemical hygiene practices. Wash contaminated clothing before reuse. When using do not eat, drink or smoke. Always remove oil with soap and water or skin cleaning agent, never use organic solvents. Do not use oil-contaminated clothing or shoes, and do not put rags moistened with oil into pockets.

SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

9.1. **Information on basic physical and chemical properties**

Appearance:	Clear
Colour:	Light & pale yellow.
Odour:	Mild Hydrocarbon odor.
Odour threshold:	Not available.
pH:	Not applicable.
Melting point / freezing point:	Not available.
Boiling point:	Not available.
Flash point:	170°C ASTM D 92
Evaporation rate:	Not available.
Flammability (solid, gas):	Not applicable.
Explosion limits:	Not available.
Vapour pressure:	Not available.
Vapour density:	Not available.
Relative density:	0.80 - 0.90 (15°C)
Solubility:	Insoluble in water.
Partition coefficient (n-octanol/water):	Not available.
Auto-ignition temperature (°C):	Not available.
Decomposition temperature (°C):	Not available.
Viscosity:	68 cSt (40°C)
Explosive properties:	Not available.
Oxidising properties:	Not available.

9.2. **Other information**

Other data:	Pour point (°C): -36

SECTION 10: STABILITY AND REACTIVITY

10.1. **Reactivity**

Reactivity: Not reactive.

10.2. **Chemical stability**

Stability: Stable under normal temperature conditions.

10.3. **Possibility of hazardous reactions**

Hazardous reactions: None known.

10.4. Conditions to avoid

Conditions/materials to avoid: Heat, sparks, flames.

10.5. **Incompatible materials**

Incompatible materials: Strong oxidizing substances.

10.6. **Hazardous decomposition products**

Hazardous decomposition products: None under normal conditions. Hazardous polymerisation will not occur.

SECTION 11: TOXICOLOGICAL INFORMATION

11.1. Information on toxicological effects

Acute Toxicity (Oral):	Based on available data, the classification criteria are not met.
Acute Toxicity (Dermal):	Based on available data, the classification criteria are not met.
Acute Toxicity (Inhalation):	Based on available data, the classification criteria are not met.
Skin Corrosion/Irritation:	Based on available data, the classification criteria are not met.
Serious eye damage/irritation:	Based on available data, the classification criteria are not met.
Respiratory or skin sensitisation:	Based on available data, the classification criteria are not met
Germ cell mutagenicity:	Based on available data, the classification criteria are not met
Carcinogenicity:	Based on available data, the classification criteria are not met
Reproductive Toxicity:	Based on available data, the classification criteria are not met
STOT - Single exposure:	Based on available data, the classification criteria are not met
STOT - Repeated exposure:	Based on available data, the classification criteria are not met
Aspiration hazard:	Based on available data, the classification criteria are not met

Inhalation: Inhalation of oil mist or vapours formed during heating of the product may irritate the respiratory. system and provoke throat irritation and coughing.

Skin contact: Prolonged or repeated contact with used oil may cause serious skin diseases, such as dermatitis.

Eye contact: May cause temporary eye irritation.

Ingestion: Low order of acute toxicity, but aspiration following ingestion and vomiting may cause severe and potentially fatal chemical pneumonitis .

Specific effects: Used oil may contain harmful contaminants.

SECTION 12: ECOLOGICAL INFORMATION

12.1. **Toxicity**

Ecotoxicity: Not harmful to aquatic organisms. Expected LC/EC50 value > 100 mg/l

12.2. **Persistence and degradability**

Degradability: Potentially degradable, but will persist in the environment for long periods.

12.3. **Bioaccumulative potential**

Bioaccumulative potential: No data available on bioaccumulation.

12.4. **Mobility in soil**

Mobility: Spillages may penetrate the soil causing ground water contamination

12.5. **Results of PBT and vPvB assessment**

PBT/vPvB: Not known.

12.6. **Other adverse effects**

Not known.

Other adverse effects:

SECTION 13: DISPOSAL CONSIDERATIONS

13.1. Waste treatment methods

Waste is classified as hazardous waste. Dispose of waste and residues in accordance with local authority requirements.

Contaminated packaging: Disposal through approved facilities is recommended.

SECTION 14: TRANSPORT INFORMATION

The product is not regulated as dangerous goods under IMDG Code, IATA-DGR and ADR/RID.

	DOT	TDG	IATA	IMDG/IMO
UN Number	Not regulated	Not regulated	Not regulated	Not regulated
Un proper Shipping Name	Not regulated	Not regulated	Not regulated	Not regulated
Transport hazards Class	Not regulated	Not regulated	Not regulated	Not regulated
Packing Group	Not regulated	Not regulated	Not regulated	Not regulated
Environmental hazards	No	No	No	No
Transport in bulk according to Annex II of MARPOL 73/78 and the IBC Code	Not regulated	Not regulated	Not regulated	Not regulated

14.7. **Transport in bulk according to Annex II of MARPOL 73/78 and the IBC code**
Transport in bulk: Not regulated.

SECTION 15: OTHER INFORMATION

The user must be instructed in the proper work procedure and be familiar with the contents of these instructions.

Handling of used oils:
Protect health - avoid prolonged and repeated skin contact. Wash with soap and water. Protect the environment - do not pollute drains, water courses or the soil. Contact your local authority for any used oil disposal instructions.

The information on this data sheet represents our current data and is reliable provided that the product is used under the prescribed conditions and in accordance with the application specified on the packaging and/or in the technical guidance literature.